

Introduction to Artificial Intelligence

Types of AI — Narrow AI, General AI and Super AI; Weak AI vs Strong AI

Classification by Scope/Capability

Narrow AI (Artificial Narrow Intelligence, ANI)

- Designed and trained to perform a single, specific task or a narrow range of related tasks.
- All AI systems in use today fall into this category — voice assistants, recommendation engines, facial recognition, spam filters, chess-playing programs.
- Cannot transfer its competence to a task outside its training domain (a chess AI cannot drive a car).

General AI (Artificial General Intelligence, AGI)

- A hypothetical form of AI with the ability to understand, learn, and apply knowledge across any intellectual task a human can perform.
- Would possess flexible, transferable reasoning and common sense across domains, not just one narrow task.
- Does not yet exist; remains an active area of research and debate.

Super AI (Artificial Superintelligence, ASI)

- A hypothetical future form of AI that would surpass human intelligence across virtually all domains — including creativity, problem-solving, and social skills.
- Purely theoretical at present; raises significant philosophical, ethical, and safety questions about control and alignment with human values.

Classification by Functionality/Depth

Weak AI

- Simulates intelligent behaviour to accomplish a specific task, without possessing genuine understanding, consciousness, or subjective experience.
- Nearly synonymous with Narrow AI in practice — e.g., Siri or Alexa appear to "understand" speech but are executing pattern-matching and statistical models.

Strong AI

- A hypothetical AI that would possess genuine understanding, consciousness, and mind — capable of actual thought, not merely its simulation.
- Closely tied to the philosophical question of machine consciousness; remains theoretical.

Consolidated View

Classification	Category	Real / Hypothetical	Example
By Scope	Narrow AI (ANI)	Real — exists today	Siri, Google Maps, spam filters
By Scope	General AI (AGI)	Hypothetical	A single system doing any human task equally well
By Scope	Super AI (ASI)	Hypothetical	Intelligence surpassing all humans in all fields
By Depth	Weak AI	Real — exists today	Chatbots, recommendation systems
By Depth	Strong AI	Hypothetical	A machine with genuine consciousness

Note the overlap: nearly every AI system in use today is both Narrow AI and Weak AI — it performs one task well through simulated intelligence, without genuine understanding or cross-domain flexibility.

Tasks Performed by AI; AI Capabilities and Limitations

Tasks Performed by AI (Russell & Norvig Classification)

Mundane Tasks

- Everyday tasks that humans perform effortlessly but are surprisingly hard for machines: perception (vision, speech), natural language use, common-sense reasoning, navigating and manipulating the physical world.
- Examples: recognising a face in a photo, understanding a spoken sentence, a robot navigating a cluttered room.

Formal Tasks

- Tasks governed by well-defined, formal rules — games and mathematical/logical problem-solving.
- Examples: playing chess or Go, proving mathematical theorems, verifying logical statements.

Expert Tasks

- Specialised tasks that require deep domain expertise typically possessed by trained human professionals.
- Examples: medical diagnosis, financial/investment analysis, engineering design, legal document review.

Interestingly, AI historically found "formal" and "expert" tasks easier to tackle than "mundane" tasks — a computer could beat a chess grandmaster before it could reliably recognise a cat in a photograph, because mundane tasks rely on tacit human skills that are hard to formalise.

AI Capabilities

- Processing and analysing massive volumes of data far faster than humans.
- Recognising complex patterns in images, sound, and text (computer vision, speech recognition, NLP).
- Automating repetitive and rule-based tasks with high consistency and minimal error.
- Making predictions and recommendations based on historical data (forecasting, personalisation).
- Operating continuously without fatigue, enabling round-the-clock services.
- Optimising complex systems (logistics routing, resource allocation, energy grids).

AI Limitations

- Lack of genuine common-sense reasoning and contextual understanding.
- Dependence on the quality and quantity of training data — "garbage in, garbage out."
- Susceptibility to bias — models can learn and amplify biases present in training data.
- Poor generalisation to situations far outside their training distribution.
- Limited explainability — many advanced models (especially deep learning) act as "black boxes."
- No genuine creativity, emotion, or self-awareness — outputs are statistical recombination of learned patterns.
- High computational and energy costs for training and deploying large models.
- Ethical and safety concerns: privacy, accountability, misuse (e.g., deepfakes), and job displacement.

AI Myths vs Realities; AI in Everyday Life; AI vs Automation; Interdisciplinary Scope of AI

AI Myths vs Realities

Myth	Reality
AI is fully autonomous and thinks like a human.	Today's AI (Narrow/Weak AI) performs specific tasks using statistical patterns; it does not "think" or possess consciousness.
AI will imminently surpass and replace all human intelligence.	General/Super AI remain hypothetical; current systems lack common sense and cross-domain flexibility.
AI is always accurate and objective.	AI can be biased, wrong, and reflects flaws present in its training data and design choices.
AI will inevitably take away all human jobs.	AI automates specific tasks, often augmenting human roles and creating new job categories, rather than eliminating employment outright.
AI systems understand what they are doing.	Most AI systems perform pattern recognition and prediction without genuine comprehension or intent.
Only tech companies use AI.	AI is applied across nearly every sector — healthcare, agriculture, law, marketing, manufacturing, and government.

AI in Everyday Life

- Smartphones: voice assistants (Siri, Google Assistant), predictive text, face-unlock.
- Streaming and shopping: personalised recommendations (Netflix, YouTube, e-commerce platforms).
- Navigation: real-time route optimisation and traffic prediction (Google Maps).
- Banking: fraud detection, credit scoring, chatbots for customer service.
- Social media: content feeds, spam/toxicity filtering, image tagging.
- Home devices: smart thermostats, robotic vacuum cleaners, smart speakers.

AI vs Automation

Aspect	Traditional Automation	Artificial Intelligence
Basis	Fixed, pre-programmed rules/instructions	Learns patterns from data; can adapt behaviour
Flexibility	Performs the same task the same way every time	Can adjust decisions based on new data or context
Decision-making	Follows a rigid, predefined sequence	Can weigh probabilities and make predictive judgments
Example	An assembly-line robot arm following fixed motions	A recommendation engine adapting to changing user behaviour

In short, automation executes; AI decides. All AI-driven systems automate tasks, but not all automation involves AI — many automated systems simply follow static, hand-coded instructions with no learning or adaptation.

Interdisciplinary Scope of AI

- **Healthcare:** Medical image analysis (radiology), disease diagnosis support, drug discovery, personalised treatment planning, hospital resource optimisation.
- **Business:** Demand forecasting, customer segmentation, supply-chain optimisation, algorithmic trading, HR analytics.
- **Law:** Contract review and analysis, legal research assistance, predictive analytics for case outcomes, e-discovery.
- **Marketing:** Personalised advertising, customer sentiment analysis, chatbots for engagement, campaign optimisation.

This interdisciplinary reach illustrates that AI is not a narrow, isolated technical discipline but a general-purpose toolkit whose value depends on close collaboration between computer scientists and domain experts in each field.